

General Industries Hub

Enterprise Inventory Management

Logistics Portal Manual

Purchase orders, vendor management, procurement chain.

Version 1.2.0 - Generated 2026-10-03

Confidential - for authorized personnel only

GI Hub - Role booklet - Logistics Portal Manual

Contents

1. Introduction & System Overview	3
1.1 What the system does	3
1.2 Core principles	3
1.3 The transaction lifecycle (the heart of the system)	4
1.4 Currency, dates, units	5
3. Login, Sidebar & Common Elements	6
3.1 Login screen	6
3.2 Sidebar (visible after login)	7
3.3 Brand header (top of every page)	7
3.4 Bug / Feature reporting dialog	7
3.6 Notifications bell (NEW in v3.0)	8
3.5 Overdue Returnable banner (Store Keepers only)	10
3.7 Saving an entry when the signal drops	10
3.8 What's new	10
3.9 Stock vs the Excel workbook - which materials differ, and why	11
3.10 Lots, FEFO and the Lot Register workbook	12
3.11 What changed in September-October 2026 (Phases 14, 15 and 16)	15
3.12 The Hub Assistant answers some questions directly (Phase 17, October 2026)	18
11. Status Codes, Reason Codes & Glossary	20
11.1 Common status pills	20
11.2 Adjustment reason codes	20
11.3 PR workflow states	21
11.4 Lot statuses	21
11.4a DN states (NEW v3.0)	21
11.4b PO + PO line states (NEW v3.0)	21
11.4c Force-closure target types (NEW v3.0)	21
11.4d Reschedule + vendor return states (NEW v3.0)	21
11.4e RL/BL family tags (NEW v3.0)	21
11.4f Notification severity (NEW v3.0)	22
11.4g Logistics-status on PR rows (NEW v3.0)	22
11.5 Glossary	22
14. Logistics Portal Manual	24
14.1 Pages visible to Logistics	25
14.2 Tab 1: Incoming PRs	25
14.3 Tab 2: Create PO	26
14.4 Tab 3: Open POs	27
14.5 Tab 4: Assign to Warehouse	28

14.6 Tab 5: Reschedules	28
14.7 Tab 6: Force-Close	29
14.8 Tab 7: <- Vendor Returns	29
14.9 Tab 8: History	30
14.10 Logistics - Use Cases	30
14.11 Logistics - FAQ	31
16. Cross-Role Procurement Walk-through	32
16.1 The scenario	32
16.2 Hour by hour	32
16.3 Side-path examples	34
16.4 What stops a duplicate PR or a double order	35
22. Quality, Safety, Employees & Procurement (QSEP)	37
22.1 Quality Control - the new role and the issue block	37
22.2 PPE - issued through the ordinary form	40
22.3 Employees - one person, one number	41
22.4 Procurement - drafted delivery notes, urgency, and scanned paperwork	41
22.5 Assets - one hammer, one row, anywhere in the company	43
22.6 Passwords	43
26. Practice Mode - learn the system without touching Live	44
26.1 Switching between Live and Practice	44
26.2 How you know you are in Practice	44
26.3 The practice accounts	45
26.4 What is in Practice	45
26.5 What works differently in Practice	46
26.6 Offline entries stay where they were made	46
26.7 Resetting Practice (admins)	46

1. Introduction & System Overview

1.1 What the system does

The General Industries (GI) Hub is a multi-site warehouse inventory, procurement and asset-management system. It tracks every material movement - issues, receipts, returns and corrections - through a two-stage approval process, and gives you live stock figures, oldest-stock-first discipline on anything with an expiry date, stock valuation, serialised tool and equipment tracking, and a complete audit trail across every site you manage.

It runs in a web browser on a phone, a tablet or a desktop, and is also available as an installable app for Windows, macOS and Android.

1.2 Core principles

Principle	What it means
Stock is calculated, never stored	Stock on hand is always received, minus consumed, minus returned, recalculated from the movement history every time it is shown. No running total is kept anywhere, so no running total can drift out of step with reality.
Two-stage approval	Store Keepers stage entries -> HOD reviews and commits at End-of-Day. Nothing touches the permanent ledger without HOD approval.
Site isolation	HODs and Supervisors see only their own site's stock. Only Admin sees all sites. Cross-site moves require formal request + approval.
Audit-first	Every consequential action is recorded with the user's name, the exact time and what changed. Even a deletion leaves a trace.
Nothing to maintain	Upgrades apply themselves when the system starts. Nobody has to run a database update by hand.
Procurement chain	One continuous workflow from a site's request through to material on the shelf. Logistics owns purchase orders; Warehouse owns physical receiving and delivery notes; the Site HOD approves what is being sent; the Store Keeper confirms it arrived. Every handover is recorded.
RL/BL strict separation	Rubber Lining and Brick Lining items NEVER share a PO line group, a DN, or a warehouse aggregation. The system rejects mixed-family DNs by design and tags each line with its family on insert.

Principle	What it means
Warehouse-blind pricing	Warehouse users can see materials and quantities but NEVER see Unit_Price, Total_Price, or any monetary header field on a PO. Three independent enforcement layers guarantee this.

1.3 The transaction lifecycle (the heart of the system)

Every stock movement travels the same four steps. Nothing reaches the permanent record until a Head of Department has approved it.

Step	Who does it	What happens	Where it sits afterwards
1	Store Keeper	Enters the issue, receipt or return on the Entry Log	Staged, awaiting approval
2	Head of Department	Reviews it at End-of-Day and may edit the quantity, approve, or reject with a reason	Still staged until they decide
3	System	Commits the approved entry to the permanent stock record	Permanent - stock figures move now
4	System	Sends the WhatsApp and in-app notifications, and writes the audit entry	Permanent audit history

Receipts, issues, returns and stock adjustments all follow this same shape. A rejected entry never reaches step 3, so it never affects a stock figure - but the rejection itself is still recorded, together with the reason.

1.3a The procurement chain

Buying material is a relay. Each role does one thing and hands over to the next, and every handover appears in the receiving role's own queue.

Step	Role	Action	Lands in
1	Site HOD	Submits a Purchase Request	Logistics - Incoming PRs
2	Logistics	Issues the Purchase Order	Site HOD is notified; Admin sees it for oversight
3	Logistics	Assigns the order (or individual lines) to a warehouse	Warehouse - Incoming Assignments
4	Warehouse	Acknowledges, then	Warehouse - Receiving

Step	Role	Action	Lands in
		records the physical arrival from the vendor	
5	Warehouse	Prepares the Delivery Note	Logistics - DN approval queue
6	Logistics	Approves the delivery date	Site HOD - DN Approvals
7	Site HOD	Approves the contents	Store Keeper - Incoming DNs
8	Store Keeper	Confirms physical arrival on site	Permanent stock record; the delivery closes

Three side-paths exist for when things do not go to plan:

- Vendor returns - any role can raise one. The order line reopens so the material can be re-delivered.
- Reschedules - Warehouse or Site HOD request a new date; Logistics decides.
- Force-closure - Logistics only, for an order that will never complete. Both Admin and the Site HOD are notified automatically.

A Delivery Note may never mix Rubber Lining and Brick Lining material. The system refuses to create one, rather than warning about it.

1.4 Currency, dates, units

- Currency: Saudi Riyal (SAR) throughout.
- Dates: shown as DD/MM/YYYY or DD MMM YYYY.
- Time zone: the site's local time. Audit entries record the time to the second.
- Units of measure: each item carries its own unit - pieces, boxes, rolls, cans, metres, kilograms. There is no automatic conversion: an item is issued in the same unit it was received in. If a material arrives in boxes and is used in pieces, record the conversion in the item's description and keep one unit for both directions.

3. Login, Sidebar & Common Elements

3.1 Login screen

Elements:

- Live | Practice switch - above the username. Live is the real system and is selected unless you choose otherwise. Practice is a separate training copy with made-up data, where nothing you do affects Live. Switching signs you out. See §26 Practice Mode for everything about it.
- Username text box - your assigned username (case-sensitive).
- Password text box - masked input. Min 8 characters policy enforced on creation.
- Sign In button - checks your credentials and takes you straight to the first page your role can open. This works from any address, including a bookmarked /login; an address that is not a page takes you to your home page instead of a blank screen.
- "Don't have an account? Request access" link - opens a self-service registration form.

Registration form elements:

- Username - must be unique
- Password / Confirm Password
- Role requested - selectbox: Store Keeper / Supervisor / HOD (Admin role cannot be self-requested)
- Site_ID - which site to associate the account with
- Phone Number - used for WhatsApp alerts (format: +966 5X XXX XXXX)
- Submit Request button - creates a pending_users row + audit log + WhatsApp alert to all admins

Admin must approve via Admin Portal -> Users tab before the user can log in.

3.1.1 The look of the sign-in page

Phase 14e, redesigned in Phase 15c.

The large gold GI mark sits in its own space at the top of the page, and the glass sign-in card sits below it, so the two never overlap. The mark rises into place once when the page opens. After that it stands still, and a slow band of light sweeps across the gold every few seconds. Nothing on the page follows the mouse. What you see depends on your device, and the form works the same on all of them, from the first moment:

- Everyone, including the Android, iOS and desktop apps: the mark drawn in the page's style.
- A desktop computer with a graphics card: the same mark in 3D gold, in the same place. It loads separately after the page is ready, so it never delays signing in. It draws nothing between sweeps, stops when you switch tabs, and is removed as soon as you sign in.
- A very short window (under about 560 pixels tall): the large mark gives way to a small, still mark inside the card.
- Anyone with Reduce motion switched on in their system settings: no rise and no sweep. The mark simply stands there.

The Executive Summary and the Training page use the same glass style. Their cards rise gently into place as you scroll, except with Reduce motion on and when printing.

3.2 Sidebar (visible after login)

Every page shares this sidebar. Reading top to bottom:

Element	Purpose	Notes
GI Hub bolt icon + version	Branding	Shows v3.0.0
Role card	Your username + role badge	Color-coded: grey=Store Keeper, emerald=Warehouse, blue=Supervisor, indigo=HOD, sky=Logistics, gold=Admin
Notifications bell (NEW v3.0)	Unread count + inbox dialog	Red badge if N>0, primary button changes to "Open inbox (N unread)". Modal shows recent procurement events with mark-read controls. See §3.6
"Navigate to:" radio	Page picker	Only shows pages allowed by your role
INVENTORY ALERTS section	Compact stock-alert badge	Visible to Supervisor/HOD/Admin only. Shows count of items below minimum (or "All levels adequate" if clear)
Bug/Feature reporting	Opens a dialog	See §3.4
Theme toggle	Dark / Light mode switch	Persists per-session
Sign Out button	Ends the session	Audit-logs the LOGOUT event

3.3 Brand header (top of every page)

Variant	Used by	Visual
Standard header	Live Dashboard, Entry Log	Gold subtitle accent + current date
HOD header	HOD Portal	Purple subtitle accent + current date
Admin header	Admin Portal	Gold accent + green/amber pulse chip ("All systems operational" / "Degraded") + current date

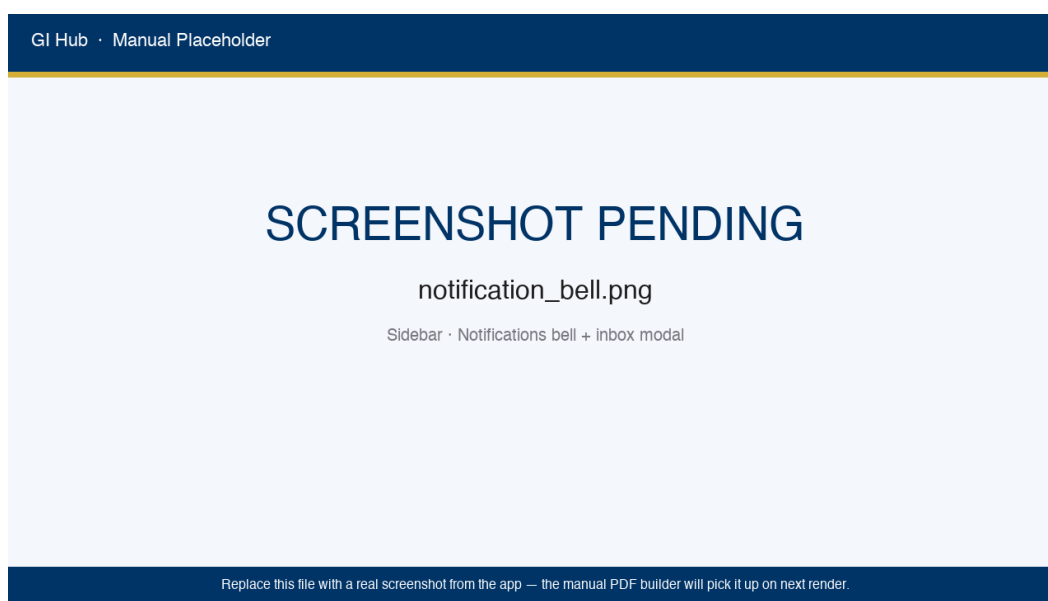
3.4 Bug / Feature reporting dialog

Available to every user from the sidebar:

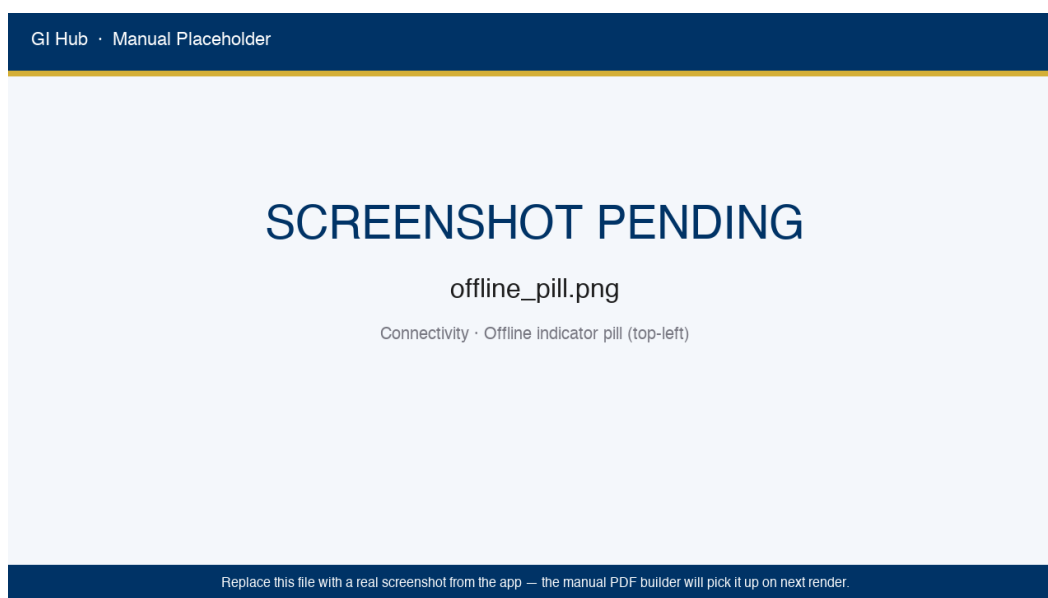
- Type selectbox - Bug Report / Feature Request
- Page dropdown - which page the issue/idea relates to (Live Dashboard, Entry Log, HOD Portal, Admin Portal, Reports, Other)
- Description textarea - up to 200 characters
- Submit button - writes to bug_reports table with user, timestamp, type, page, description. Visible to Admin in the Admin Portal Reports & Bugs tab for triage.

3.6 Notifications bell (NEW in v3.0)

Sits between the role card and the navigation radio. Every signed-in user sees their own personalised inbox of procurement-chain events.



Sidebar - Notifications bell with unread badge + inbox modal



Connectivity - offline indicator pill appears top-left when the browser loses network (Phase 7E)

The button:

- No unread: "Open inbox" (secondary button, no badge)
- 1+ unread: "Open inbox (N unread)" (primary button, red pill badge with the count, capped at "99+")

The inbox dialog (click the button):

- Only unread toggle - on by default. Flip off to see your full history.
- Per-notification card - colour-coded left border by severity:
 - Critical (red) - force-closures, T-0 delivery reminders
 - Warning (amber) - T-1/T-2 reminders, reschedule requests
 - Success (green) - DN approved / received successfully
 - Info (blue) - PR submitted, PO issued, assignments
- Per-row Mark read - flips just that notification
- Bulk [OK] Mark all as read - flips every visible row (respecting the role + site + warehouse scope you'd see normally)

What gets sent here:

Event you'll see	Triggered by
New PR from a site (Logistics only)	Site HOD pressing " Submit PR(s) to Logistics"
PO issued (Site HOD)	Logistics creating a PO against your site's PR
PO assigned to warehouse (Warehouse only)	Logistics routing a PO to your WH
DN awaiting your approval (HOD)	Logistics approving a DN delivery date
Incoming DN ready to receive (SK)	HOD approving DN content
Reschedule requested (Logistics)	Warehouse or HOD asking to push a date
Reschedule decided (requester)	Logistics approve/reject
Force-closure (Admin + originating HOD)	Logistics force-closing a PR/PO/line

Event you'll see	Triggered by
Vendor return raised (Logistics)	Any role raising a return
Delivery reminder T-2 / T-1 / T-0	The daily sweep job (see §9)

The bell is tolerant - if the notification lookup fails, the badge silently shows 0 instead of breaking the sidebar. In-app notifications ALWAYS fire; the WhatsApp side is gated by the per-event WhatsApp toggles in the system settings (see §9).

3.5 Overdue Returnable banner (Store Keepers only)

When a Store Keeper logs in and there are overdue returnable items at their site, a red banner appears at the top of any page they navigate to:

> [!] OVERDUE ITEMS - Action Required: N borrowed item(s) past expected return: <item names>. Go to the Returnable Items tab to follow up.

3.7 Saving an entry when the signal drops

If the network drops while you save a receipt, issue, return, stock adjustment or bulk batch, GI Hub keeps the entry on your device and shows a saved offline message instead of an error. A badge in the header shows how many entries are waiting. They are sent by themselves when the connection returns, and you can press the badge to send them straight away.

An entry is never saved twice. Each save carries its own reference, and the server recognises it if it arrives again. That can happen when the page reloads while the entry is being sent, or when the connection drops just as the server confirms. The second arrival is matched to the first, and only one entry reaches your HOD. You never need to check for, or delete, a duplicate caused by a bad connection.

3.8 What's new

Phase 14d.

When a feature you use changes, a What's new panel opens by itself the next time you sign in. It shows only what concerns you: an announcement is sent to the roles that can open the page it is about, so nobody is told about a screen they cannot reach.

Each card says what changed. Open it takes you to the page, Watch the tutorial plays the short video when there is one, and a User Manual § tag names the section to read.

Press Got it and the panel closes. It does not open by itself again for the same announcements. To see recent ones, press the gift icon in the top bar, next to the bell; a number on it means something new is waiting.

A matching notification also appears in the bell (§3.6). Announcements are in-app only - never WhatsApp or email.

3.9 Stock vs the Excel workbook - which materials differ, and why

Added 26 September 2026.

GI Hub keeps its own stock ledger, and the site also keeps CNCEC_Inventory.xlsx. After every Excel sync, GI Hub compares its stock for every material with the workbook's Current Stock column. It then works out why any material differs.

Where: the top of the Stock page.

- A yellow banner says how many materials differ, for example 14 of 505 materials don't match the Excel workbook, and when that was checked.
- Each of those materials has a red row and a red != Excel tag. Point at the tag to see the reason.
- Review the differences lists every material. Each one shows its workbook and GI Hub figures, what caused the difference, where to look (a GI Hub record number or an Excel sheet and row) and how to fix it.
- A material's own page (scan its QR or open it from Stock) shows the same warning.

What the causes mean:

Tag	What happened	How to fix it
Only in GI Hub	A receipt, issue or return was entered in GI Hub, and the workbook's log never got it. The entry names who made it and its remarks.	If it really happened, add it to the log. If it was a test or a mistake, reverse it in GI Hub with a Stock Adjustment, or have an admin remove it.
No longer in Excel	GI Hub took a line from the workbook, and the workbook has since lost it (deleted or re-dated).	Put the line back if that was a mistake. Otherwise an admin removes it with the sync's --prune-vanished option.
Not synced yet	The workbook has a line GI Hub doesn't have yet.	Run the Excel sync.
Edited in Excel	A quantity was changed in the workbook after the last sync.	Run the Excel sync.
Sheet totals stale	The Inventory sheet's own Current Stock doesn't match its own logs. Current Stock is a typed value in this workbook, so it can fall behind.	Correct the figure on the Inventory sheet.
Opening stock	The Opening Stock differs.	Run the sync, or correct the workbook.
Unexplained	Part of the difference that none of the above explains. It is always shown, never hidden.	Compare the material's records by hand.

The marked workbook. Press Get the marked workbook and choose your CNCEC_Inventory.xlsx. You get back a copy containing:

- a GI Hub check sheet first, listing every cause with how to fix it;
- each differing material's row on the Inventory sheet filled red, with a note on its Current Stock cell;
- any log line the check names filled amber.

Make the fixes in your own workbook, then sync again. Your file itself is never changed. Check again uploads the workbook and refreshes the list without downloading anything.

Store Keepers, Warehouse, HODs, Logistics and Admins can upload. Everyone who can see Stock can read the result. Nothing on this screen changes stock.

[!] The check only explains. It never corrects stock by itself, because deciding whether an entry was real is a person's call.

3.10 Lots, FEFO and the Lot Register workbook

Phase 16 (October 2026).

A lot is one batch of a material: the cans of PU COMP A from batch 3504, or the CHEMOLINE rolls of batch 1O25003382. GI Hub keeps the stock of each lot, so it can tell the store keeper which lot to use first (FEFO, first expire, first out).

3.10.1 What the `Serial No.` column means in the workbooks

The Receipt, Consumption and Return Logs have one Serial No. column. GI Hub reads it according to the item on that row:

The item is...	Serial No. is...	GI Hub keeps it as...
a Surface Shield (PU, Furan, Phenacin, Coroflake, BC 3004...)	the batch	the Lot No. (and the cell as typed)
a roll (CHEMOLINE, UOM ROL)	the roll number	the roll number, and its batch as the Lot No.
a brick (AR 30/40 MM, carbon bricks)	not a lot - bricks have no batch and never expire	the serial, with no lot
anything else (pumps, tools, equipment)	the equipment / asset no.	the serial, as before

Rules that keep this safe:

- A lot is spelled one way. 3504.0 is 3504, and A4525 and A 4525 are the same lot (A 4525). N/A or a blank cell means no lot.
- Roll numbers begin with 1O (the letter O). A roll typed 10... (with a zero) is corrected to 1O..., and the sync says so.
- One lot per cell. A cell such as 4525 = 55 Cans; 1823 = 2 Cans gets no lot. Split it into one row per lot in the workbook; the sync names the rows that need it.
- A lot that was never received is reported, not invented. If the Consumption Log names a lot no receipt brought in for that material, the sync lists it with the material it was received under (often a typing slip), and the consumption still counts toward stock.

- History keeps the lot it was given. GI Hub never picks a lot for an old workbook row that has none. Automatic FEFO applies only to issues entered in the app.
- A return gives back to its lot. The Return Log's Serial No. is now read too: a Surface Shield return carries its lot, and an equipment return keeps its asset tag.

Every receipt that names a lot creates that lot in GI Hub. A lot's stock is always received - consumed - returned, worked out from the ledger.

3.10.2 The Lot Register workbook ('Rubber & Brick Materials - CNCEC.xlsx')

This workbook keeps, per Surface Shield material, one sheet of what arrived: batch, manufacture date (MFD), expiry, DN No. and quantity. Put it in the same folder as the other workbooks. The normal sync reads it automatically; any file named like *Rubber*Brick*CNCEC*.xlsx is found, and the run prints a > lots section.

[!] It never changes stock. Stock comes only from the Receipt, Consumption and Return Logs. This workbook only describes the lots GI Hub already has: their MFD, expiry and batch. Its quantities are compared with the Receipt Log, and every difference is reported. A mistake in it can make an expiry wrong, but it can never double a quantity.

How each row is matched:

1. To its material - by the SAP column. If a sheet has no SAP column, GI Hub matches the material name, the component letter (COMP C) and the pack size (10 Kg vs 2.5 Kg), and refuses anything it cannot pin to exactly one item. The material code is only a cross-check, because COMP A-D share one code.
2. To its receipt - the same SAP, DN No. and received date. CHEMOLINE rolls have no DN, so they match on SAP and date, roll by roll.
3. To its lot - the Batch No. (normalised as in §3.10.1). For CHEMOLINE the Order No. is the batch and each Roll No. is a roll inside it.

What it fills in:

On...	It writes...
the lot	MFD, expiry, batch reference, DN No.
the roll register	each CHEMOLINE roll and the batch it belongs to
a receipt with no lot	the lot, when the workbook names exactly one batch for it
an item with no shelf life	the shelf life (months), learned from its own MFD -> expiry pairs

Expiry:

- An expiry in the workbook is used as it is (from the file).
- A missing expiry is MFD + the item's shelf life (derived).
- If two rows of one lot disagree, the earliest expiry is used.
- An expiry before its MFD is refused and reported.
- An expiry a person typed on the Receive form is never overwritten.

Bricks are compared with the receipts but get no lot (they have no batch and never expire).

What the sync tells you (each is listed by sheet and row):

- quantities that differ from the Receipt Log;
- rows with no matching receipt;
- expiries before the MFD;
- lots whose rows disagree on dates;
- material codes that disagree with the inventory;
- a batch the Receipt Log spells differently (e.g. 0926 vs 0.926);
- rows with no batch.

3.10.3 Lots & Expiry - which lot to use, and what is about to expire

The Lots & Expiry page (sidebar, under Stock) is for the Store Keeper, QC, the HOD and the Head of Qualities. It lists every lot of a lot-tracked material, the oldest expiry first, with:

- MFD and expiry - an expiry marked derived is MFD + the item's shelf life;
- status - Expired, <= 30 days, <= 60 days, <= 90 days, OK, No expiry, or Quarantined / Disposed when the Admin has taken the lot out of use (Admin Console -> Lots). A quarantined or disposed lot is never offered on the Issue form;
- received, consumed, returned and remaining - worked out from the ledger.

Click a status at the top to show only those lots. Show used-up lots adds the lots with nothing left. A CHEMOLINE batch opens to show its rolls: when each was received, where it is, and for a used roll the day and tank it was issued to. A red banner warns when an expired lot still has stock. At the bottom, Lots used but never received lists lots the Consumption or Return Log names that no receipt brought in.

Issuing (the Issue form's Lot field). For a lot-tracked material, the Lot field is a list of that material's open lots in FEFO order. Each shows its expiry, the days left and the quantity left.

- The first one is marked FEFO. Leaving the field blank issues from it, because GI Hub's own automatic pick uses the same order.
- [!] An expired lot is listed last, marked red, and is never the suggestion, even though its date is the earliest. You can still choose it; FEFO warns, it never blocks.
- Choosing any lot other than the FEFO lot asks for a reason, which goes to the HOD with the issue. Choosing the FEFO lot asks nothing.
- CHEMOLINE: pick the roll; its batch fills in by itself. A roll already issued is no longer offered.

Receiving (the Receive form). For a lot-tracked material the form asks for the batch (as printed on the label), the manufacture date (MFD) and the expiry:

- If you leave the expiry blank, it becomes MFD + the item's shelf life, and the form shows the date it will use.
- An expiry before the MFD, or an MFD in the future, is refused.
- An expiry you type is kept, and the Lot Register workbook never overwrites it.

The daily expiry notice. Each evening the Store Keeper and the HOD of a site get one notice listing that site's lots that expire within 30 days or have already expired, each with how much is left. It is one notice per site, never one per lot. On WhatsApp it arrives inside the evening summary.

Shelf life and lot tracking (item editor). Admin -> Inventory or Records -> Inventory -> Edit has two

settings:

- Lot tracking: Automatic - every Surface Shield except bricks - or Tracked by lot, or Not tracked.
- Shelf life (months): the number a missing expiry is derived from. The Lot Register workbook fills it in where it is empty, from the item's own MFD -> expiry pairs (PU 9, Phenacin 6, BC 3004 24...).

Reading the two warnings - they are different things.

You see...	It means...	What to do
a red Expired lot with stock left	The lot was received, its expiry has passed, and the ledger says some of it is still on the shelf.	Check the shelf. If it is there, ask the Admin to dispose of it or quarantine it. If it is not, an issue or return was never entered - enter it, or correct the workbook.
a lot under Lots used but never received	The Consumption or Return Log names a lot that no receipt of that material brought in. The lot itself may be fine; usually the lot number or the SAP code on that line was typed wrong. The column Received under SAP shows the material that did receive a lot with that number.	Correct the line in the workbook (the lot, or the SAP code) and sync again. The stock already counts the line either way.

An expired lot whose stock is all used shows Used up, not Expired, and appears only with Show used-up lots switched on. So when a lot shows received 1, consumed 1, remaining 0, there is nothing to dispose of.

3.10.4 Practising with lots

Practice (\$26) has lots to try all of this on:

- PRACTICE PU PRIMER (899971) has three lots: PR-OLD (expired, still has stock), PR-SOON (expires within 30 days, so it is the FEFO suggestion) and PR-LATE.
- PRACTICE CHEMOLINE (899973) has batch 1O26009999 with three rolls to pick.
- One issue names lot PR-TYPO, which no receipt brought in, so the Lots & Expiry page has a used but never received example.

3.11 What changed in September-October 2026 (Phases 14, 15 and 16)

A one-page list for everyone who used GI Hub before September 2026. Each line names who it affects and where it is explained in full.

3.11.1 Phase 14 - quantities, reconciliation and announcements (26 September)

Change	Who	Read
Surface Shield quantities are shown both ways, kilograms first (189 KG - 21 Can). You still enter packs; a consumption form may be written in KG and is converted.	Store Keeper, Supervisor, HOD	§4.11, §4.10.8
When a QR form and the Excel workbook both record the same drum, the ledger keeps one quantity (the larger), never both.	HOD	§6.17
The Surface Shield queue is grouped: one job = one tank on one day, area asked once and credited once.	Store Keeper, HOD	§4.9a.2
What's new notices after an update, and tutorial freshness warnings for admins.	Everyone; Admin	§3.8, §7.14
The new 3D login (a still crest on phones and older computers).	Everyone	§3.1
Stock vs the Excel workbook: the Stock page marks materials whose stock differs from the workbook's, and says why.	Store Keeper, HOD, Admin	§3.9
An entry saved while the signal dropped is sent once, however many times it is retried.	Store Keeper	§3.7

3.11.2 Phase 15 - Practice, Garnet and the store keeper's note (28-30 September)

Change	Who	Read
Practice is violet with a pulsing badge, and refuses to start on an out-of-date database (no more half-working pages).	Everyone	§26.2, §17.9
Garnet is surface preparation: it credits no area of its own; the HOD answers Old or New surface per job.	Store Keeper, HOD	§4.9a.7
The store keeper's note fills the job card: Floor - 13.37 SQM Done fills the area, the part and the remark. Two notes on one tank and day are two jobs. The remark goes to the	Store Keeper, HOD	§4.9a.2

Change	Who	Read
HOD, the job history and the exports.		
The HOD adds and edits inventory items for their own site (Records -> Inventory -> New item / Edit). Deleting stays Admin-only.	HOD	§2.3.4, §2.3.8-2.3.9 (Adding an inventory item)
A user who belongs to one site no longer sees a Site box on the forms: their site is filled in and shown, not asked. Admins and multi-site users still choose.	Store Keeper, HOD	this section
The WBS No. field appears on Receive and Issue. If the HOD has not set up any WBS numbers yet it shows "None set up" and the entry still saves.	Store Keeper, HOD	§13.14, §16.5
Site and Category on a new item are matched to the spelling already in use (cncec -> CNCEC).	Admin, HOD	§2.3.8-2.3.9 (Adding an inventory item)

3.11.3 Phase 16 - lots, FEFO and the Lot Register workbook (1 October)

Change	Who	Read
The workbooks' Serial No. is read by the item: the batch for a Surface Shield, the roll for CHEMOLINE, the asset tag for equipment. Live now has real lots.	Everyone	§3.10.1
The Lot Register workbook (Rubber & Brick Materials - CNCEC.xlsx) adds each lot's MFD, expiry and rolls. It never changes stock.	Admin	§3.10.2
Lots & Expiry page: every lot, its expiry, and what is left of it.	Store Keeper, QC, HOD, Head of Qualities	§3.10.3
The Issue form's Lot field lists lots in FEFO order; an expired lot is listed last and never suggested. Choosing another lot asks for a reason (it never blocks).	Store Keeper	§3.10.3
The Receive form asks for the MFD; a blank expiry becomes MFD + shelf life.	Store Keeper	§3.10.3

Change	Who	Read
One evening expiry notice per site lists lots expiring within 30 days.	Store Keeper, HOD	§3.10.3
A return gives back to its lot, so a lot's balance is received - consumed - returned.	Everyone	§3.10.1
Practice has example lots: an expired one, a FEFO one, CHEMOLINE rolls, and a lot used but never received.	Everyone (training)	§3.10.4

3.12 The Hub Assistant answers some questions directly (Phase 17, October 2026)

The Hub Assistant (the round button at the bottom right of every page) now takes a quick look at your question before answering it - about half a second - and decides what kind of question it is. Four things can happen:

You ask...	You get
to open a page - "open the lots page", "go to Reports"	An Open ... button for that page. Only pages your role can open are ever offered; a page you cannot open, or a name that fits several pages, gets an ordinary answer instead. It only opens pages - it never does anything for you.
for a video - "is there a video on staging a return?"	The Watch it button, straight to the moment in the training video that shows it. If no video covers it, you get an ordinary written answer.
for numbers or records - "show me the receipts from last week", "which lots expire in the next 30 days?"	A small table in the chat. This is the same data as Reports -> AI -> Ask your data, and only the roles that already have that card get it: HOD, Logistics, Auditor and Admin. An HOD sees their own site only. Everyone else gets the usual written answer from the manual.
anything else - how to do something, what a word means, who can do what	A written answer from your section of the manual, exactly as before.

A question that tries to trick the assistant - telling it to ignore its rules, asking for its hidden instructions, typing database commands - is refused with the usual sentence: "I can only answer questions about your section of the manual." Ordinary work sentences that happen to use the same words ("drop the damaged drums at bay 3", "ignore the damaged drum and issue the rest") are answered normally.

If the quick check is not available (the AI service is restarting, or an admin has switched it off), the assistant simply answers as it always did. You lose nothing; only the buttons and tables do not appear.

For admins: the quick check can be switched off without a restart - Admin -> Console -> Settings, set `ai_router_enabled` to 0 (and back to 1). The AI health line shows whether it is on and ready.

Two smaller changes arrived with it:

Change	Who	Read
The Head of Qualities no longer sees Reports or the Records ledgers (Receipts, Consumption, Returns, Lots, Purchase Requests) in the menu. They never worked for this role - every one showed an error - and the role's pages are Quality Oversight and Lots & Expiry.	Head of Qualities	§2.2, §2.3.5
The Watch it button appears less often by mistake. A question that merely says "video" or "tutorial", or shares only a word like "many" or "into" with a video, no longer gets a link to the wrong moment.	Everyone	§24.2.3

11. Status Codes, Reason Codes & Glossary

11.1 Common status pills

Status	Color	Meaning
pending	grey	Awaiting decision
flagged	amber	Has a concern (e.g., zero stock at site)
approved	green	Approved, ready for commit
rejected	red	Soft-rejected (row preserved for audit)
committed	gold	Posted to permanent ledger
draft	grey	User can still edit
submitted	blue	Sent forward; awaiting next stage
in_progress	amber	Active (e.g., PR with supplier)
received	green	Fully delivered
open	blue	Active, has pending balance
closed	green	Fully fulfilled
OK / Low / Below Min / Empty	green/amber/orange/red	Stock vs minimum status
Expired / Critical / Warning	red/red/amber	Expiry buckets
sent	green	WhatsApp delivered

11.2 Adjustment reason codes

Code	Label	When to use
cycle_count	Cycle count correction	Routine periodic count
damaged	Damaged / unusable	Physical damage
expired_disposal	Expired - disposed	Expired and physically discarded
miscount_in	Miscount - found extra	Found more than system shows
miscount_out	Miscount - short	Found less than system shows
lost	Lost / unaccounted	Genuinely missing
theft	Suspected theft	Deliberate removal
return_to_supplier	<- Returned to supplier	Quality issue, sent back
other	Other (see notes)	Anything else (must explain in notes)

11.3 PR workflow states

- draft - manual creation, not yet sent
- submitted - sent to procurement
- approved - procurement greenlit
- in_progress - supplier engaged
- received - fully received

11.4 Lot statuses

- open - has remaining quantity, available for FEFO
- exhausted - fully consumed (auto)
- expired - past expiry date (manual or background)
- disposed - physically removed (manual)
- quarantine - held pending inspection (manual)

11.4a DN states (NEW v3.0)

draft -> pending_logistics -> logistics_approved -> pending_hod -> hod_approved -> pending_sk -> received With rejected as terminal from any pending state.

11.4b PO + PO line states (NEW v3.0)

PO header: open -> partially_delivered -> delivered -> closed / force_closed / cancelled PO item line: open -> partially_delivered -> delivered -> returned / closed / force_closed

11.4c Force-closure target types (NEW v3.0)

Code	Label
pr	Whole PR closed
po	Whole PO closed
po_item	Single line on a PO closed

11.4d Reschedule + vendor return states (NEW v3.0)

Reschedule: pending -> approved / rejected Vendor return: open -> vendor_acknowledged / resupplied / cancelled

11.4e RL/BL family tags (NEW v3.0)

Tag	Meaning	Detection rule
RL	Rubber Lining	The Material Code or Description contains "RL-", "RUBBER LINING" or "RUBBER-LINING"
BL	Brick Lining	It contains "BL-", "BRICK LINING", "BRICK-LINING" or "BRICK MATERIAL"
(none)	Neither family	The default when nothing above matches

The system works this out for you. An item is never tagged as both - if the wording matches both families, RL wins.

11.4f Notification severity (NEW v3.0)

Severity	Visual	When used
info (blue)	Info pings - PR submitted, PO issued, assignments	
warning (amber)	T-2/T-1 reminders, reschedule requests, rejections	
success (green)	DN approved successfully, delivery completed	
critical (red)	T-0 reminder, force-closures, urgent escalation	

11.4g Logistics-status on PR rows (NEW v3.0)

Code	Meaning
site_draft	HOD has the PR but hasn't submitted to Logistics yet
submitted	Sitting in Logistics queue waiting for PO issuance
in_po	A PO has been issued against this PR line
closed	PR fulfilled normally
force_closed	Logistics force-closed the PR with a reason

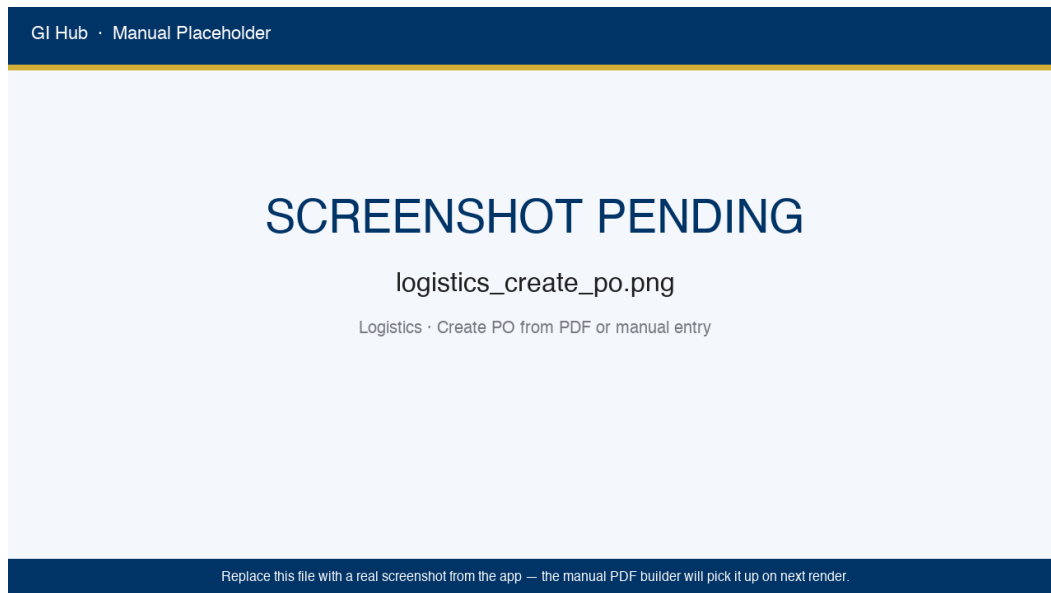
11.5 Glossary

Term	Meaning
FEFO	First-Expiry-First-Out - issue oldest-expiring lot first
EOD	End-of-Day commit (HOD action that finalizes the day's

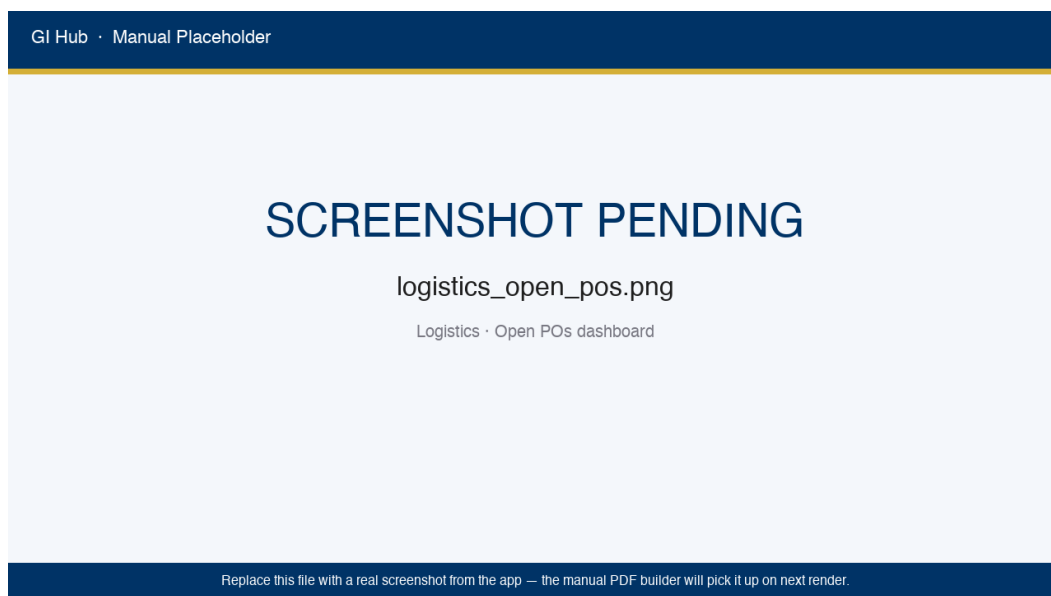
Term	Meaning
	transactions)
PR	Purchase Request
DN	Delivery Note (logistics document)
OCR	Optical Character Recognition - used for bulk-staging from images
PWA	Progressive Web App - the offline mobile companion
RBAC	Role-Based Access Control
WAL	A database setting that lets many people read while one person writes. Nothing to configure - it is on
Standard cost	Per-item Unit_Cost on inventory master (vs weighted-average cost which would require receipts-history)
Identity math	Stock derived from movements, never stored as a counter

14. Logistics Portal Manual

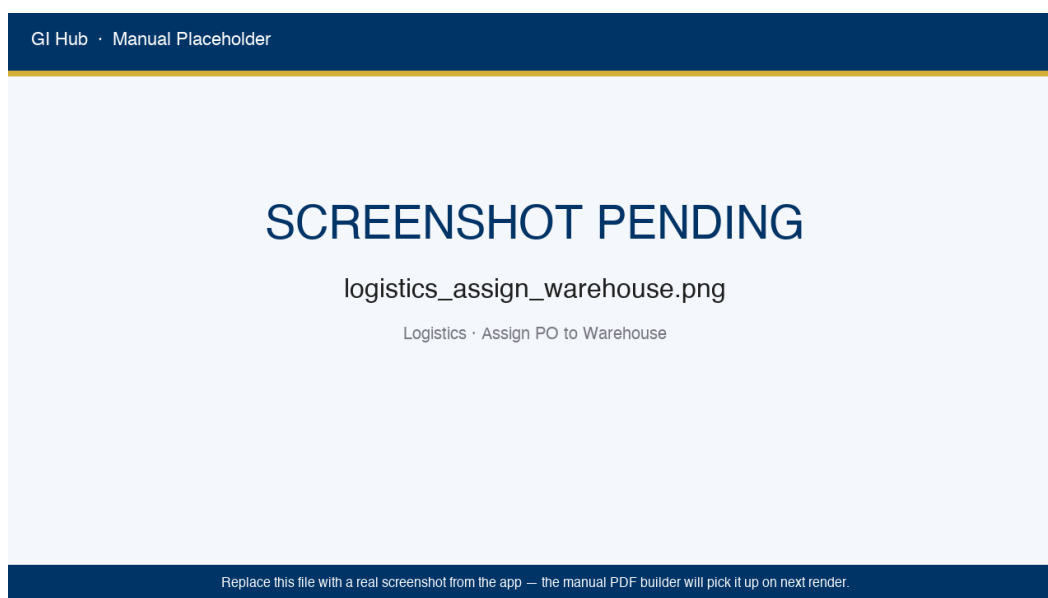
The Logistics Portal sits between Site HOD (who creates PRs) and Warehouse (which physically receives goods). Role-locked to {logistics, admin} - exact-role lock means no other role inherits access via the hierarchy. Eight tabs.



Logistics Portal - Create PO from PDF upload or manual entry



Logistics Portal - Open POs dashboard with status pills

*Logistics Portal - Assign PO to a Warehouse*

14.1 Pages visible to Logistics

- Live Dashboard (read-only - all sites)
- Logistics Portal (this page)
- Reports (incl. the 3 new procurement reports)

14.2 Tab 1: Incoming PRs

Site HODs submit PRs to Logistics from their HOD Portal -> PR tab. They land here.

14.2.1 Site filter

Dropdown of all sites + "All sites". Defaults to all.

14.2.2 Hero strip (3 cards)

Card	Source
OPEN PRs	Active queue count
TOTAL QTY	Sum across all open PR lines
SITES	Distinct sites currently submitting

14.2.3 Queue table

A table with columns: PR No. - Site - Lines - Total Qty - Submitted - Earliest Delivery - Status. Sortable, filterable, exportable.

14.2.4 Per-PR drilldown

Selectbox under the queue table. Pick a PR + Site combination and a section card appears:

Card row	Source
PR Number	the row
Site	the row
Lines	line count
Total Qty	summed across lines
Earliest Delivery	min of Delivery_Date across lines

Below: full line items grid with Material_Code, Material_Name, Requested_Qty, UOM, WBS_Number, Network, Plant, Delivery_Date, Supplier, Est_Cost_SAR.

14.2.5 Use this PR to create a PO button

Loads the PR into Tab 2 (Create PO). Selectbox state is preserved so you can switch tabs without losing the chosen PR.

14.3 Tab 2: Create PO

Two sub-tabs: Manual entry and PDF upload. Both create the Purchase Order the same way - the only difference is where the details come from.

14.3.1 Vendor picker

Selectbox of all active vendors `<code> - <name>`, plus a " Add new vendor" option that opens an inline expander with Vendor Code, Name, Address, Default Inco Terms, Default Payment Terms. Save -> re-renders the parent form with the new vendor pre-selected.

14.3.2 PO header form

Three-column layout:

Column	Fields
Left	PO Number * - PO Date - PO Type
Middle	PR Number - Quotation No. - Quotation Date
Right	Expected Delivery - Inco Terms - Payment Terms

Plus a second three-column row for Contact (vendor), Contact Email, Mobile, Our Reference, Your Reference, Our Email.

Vendor defaults (Inco / Payment Terms) auto-fill from the master row but remain editable.

14.3.3 Manual sub-tab: line items

If a PR was loaded via the Tab 1 shortcut, every PR line pre-fills with Include = True and zero Unit_Price + Total_Price (you set them). Otherwise an empty editable grid opens.

Editable columns: Include - Material_Code - Description - Qty - UOM - Unit_Price - Total_Price - WBS_Number - Network - Plant. Rows with Include = False are dropped on save.

14.3.4 PDF upload sub-tab

Element	Purpose
File uploader	Drop the PO PDF
Extract from PDF button	Reads the document and pulls out the header (PO Number, Vendor, Inco/Payment terms, Quotation refs, totals), the line items (Sr. No, Material_Code, Description, Qty, UOM, Unit_Price, Total_Price) and the PO Annexure delivery schedule (Shipment N / Material Group / Date)
Review extracted PO	Editable preview of every extracted field. The header form pre-fills. The line items grid pre-fills. Edit anything before saving.
Delivery schedule	If Annexure parsed, a table shows shipment_no - material_group - target_date
Save PO (from PDF)	Saves the PO, its line items and its shipment schedule. The original PDF is kept attached to the PO itself, so the source document is always available for audit.

On PO numbers with X-masking: the sample PDF has the last 4 digits masked as XXXX for security. The extractor preserves whatever is on the page verbatim. In production, vendors send full 10-digit numbers and those pass through unchanged.

14.3.5 Side-effects on save

- A new Purchase Order is created, open
- One line is created per item, each auto-tagged with its RL/BL family
- The matching PR lines - same PR number, same site - are marked as now covered by a PO, so they leave your Incoming PRs queue
- The site HOD gets a "PO issued" notification in-app, and on WhatsApp if that event is enabled

14.4 Tab 3: Open POs

Browse + drill into every open PO.

14.4.1 Filters

Site dropdown - Vendor dropdown - PR Number exact-match textbox.

14.4.2 KPI strip

OPEN POs count - Total value SAR - PENDING (not yet delivered) count.

14.4.3 Per-PO drilldown

Select a PO -> section card with PR Number, Vendor, Inco/Payment Terms, PO Date, Expected, Source (manual/pdf_upload), Status, Total. Below: items grid with an RL/BL family chip per row, the parsed

delivery schedule (if any), and a list of warehouse assignments.

14.5 Tab 4: Assign to Warehouse

Route a PO (or a subset of items) to a warehouse for receiving.

14.5.1 PO + Warehouse pickers

Two-column row: PO selectbox - Warehouse selectbox (shows <id> - <name>).

14.5.2 Items selector

Editable grid showing every PO line with an Include checkbox (default True). Disable lines you don't want to route - they stay with this PO for a future assignment.

14.5.3 Header

Expected Delivery date + Notes (visible to Warehouse).

14.5.4 Assign to Warehouse button

- If every line is included, the assignment is recorded as covering the whole PO - which reads more cleanly in an audit than listing each line
- If only some lines are included, the assignment records exactly which PO lines were routed
- Warehouse users at the chosen warehouse get a "PO assigned to warehouse" notification in-app, and on WhatsApp if that event is enabled
- PO header's Expected_Delivery auto-fills with this date if not set
- Audit log entry: ASSIGN_PO_TO_WAREHOUSE

14.6 Tab 5: Reschedules

Incoming reschedule requests from Warehouse / Site HOD.

14.6.1 Empty state

If no pending requests: empty-state card.

14.6.2 Per-request card

Element	Purpose
Header line	PO <number> - DN <number> - From <current_date> -> <requested_date>
Subline	requested_by + role + reason
Decide popover	Decision notes textbox + [OK] Approve / [X] Reject buttons (reject requires reason)

Approval automatically pushes the new date everywhere it matters: the PO's expected delivery date, the

warehouse assignment's expected delivery date, and the Delivery Note's date if one is linked.

14.7 Tab 6: Force-Close

Use sparingly. Force-closing notifies Admin + originating Site HOD immediately with the reason.

14.7.1 Target radio

PR (entire) - PO (entire) - PO line (single item)

14.7.2 Reason textbox

Mandatory, minimum 3 characters. Stored word-for-word on the force-closure record and in the audit log.

14.7.3 Target picker (changes per radio)

- PR: dropdown of open PR numbers in Logistics queue
- PO: dropdown of open POs
- Line: PO dropdown -> then line dropdown showing only open / partially_delivered lines

14.7.4 Force-close button

Single confirm-and-execute. Behind it:

- PR target: the PR is closed and marked force-closed in the Logistics queue
- PO target: the PO and all of its lines are marked force-closed
- Line target: only that one PO line is marked force-closed
- A force-closure record is written, carrying your reason
- Notifications go out to every Admin and to the HOD of the site the document came from, at critical severity

14.7.5 Recent force-closures (audit)

A table of the last 50 closures. Read-only.

14.8 Tab 7: <- Vendor Returns

Raise a return to the vendor against a PO. Returning a line REOPENS the PO so it shows in your active queue again.

14.8.1 PO picker + scope

PO selectbox (all POs incl. closed). Then radio: Whole PO or Single line. If single line, a line picker appears.

14.8.2 Return details form

- Return quantity (number, > 0)

- Reason (textarea, mandatory)
- Expected resupply (date picker, defaults to today + 14)
- Notes (optional)

14.8.3 <- Raise vendor return button

- A return record is created, showing it was raised by Logistics
- The PO line's returned quantity goes up, and the line reopens as either partly delivered or open
- If the PO had already been closed, it reopens as partly delivered and its closure details are cleared
- A "vendor return raised" notification goes to the Logistics inbox, and to WhatsApp if that event is enabled

14.8.4 Open returns table

A table of all open returns. Read-only.

14.9 Tab 8: History

Read-only archive. Two sub-tabs: Closed POs and Closed PRs.

- Closed POs: status in closed / force_closed / cancelled
- Closed PRs: any PR that is itself closed, or that Logistics has closed, force-closed, or already turned into a PO

For force-closures with the reason history, use Tab 6 -> Recent force-closures table.

14.10 Logistics - Use Cases

Use Case 1: Issue PO from a fresh PR

1. Incoming PRs -> pick PR -> Use this PR to create a PO
2. Create PO -> Manual entry -> pick / add vendor, fill header, edit Unit Price per line
3. Save PO -> balloon animation -> notification fires to Site HOD

Use Case 2: Issue PO from a vendor's emailed PDF

1. Receive PO PDF from vendor (sample format: Create PO -> PDF upload)
2. Drop the PDF -> Extract from PDF
3. Review extracted header + line items + delivery schedule
4. Correct anything the parser missed
5. Save PO (from PDF) - original PDF is archived

Use Case 3: Route a PO to a warehouse

1. Assign to Warehouse -> pick PO + warehouse
2. Disable lines you want to route separately later

3. Pick Expected Delivery, add a routing note
4. Assign - warehouse user is pinged

Use Case 4: Handle a Warehouse reschedule ask

1. Reschedules -> review the request (reason text + current vs requested date)
2. Decide -> notes + [OK] Approve
3. The PO's expected delivery date updates automatically; Warehouse and the requester get the decision notification

Use Case 5: Force-close a stale PR

1. Force-Close -> radio = PR (entire)
2. Write the reason (e.g. "Project cancelled by site management 2026-06-15")
3. Pick PR -> Force-close PR
4. Admin + Site HOD see a critical notification in their bell

14.11 Logistics - FAQ

Q: A PR appears in Incoming PRs but I can't issue a PO against it - Material_Code is wrong. A: PR lines come from the Site HOD's inventory catalogue. If the Material_Code is wrong, the SK created it wrong upstream. Don't fix in your PO - bounce the PR back via WhatsApp / chat, ask the site to re-submit. There's no in-app "reject PR" button (yet) because PR rejection should be a conversation, not a click.

Q: PDF extraction picked up wrong qty. A: The Review preview is editable. Always check Qty / Unit_Price / Total_Price before saving. The extractor is calibrated against the General Industries sample layout; other vendor templates may need template additions.

Q: I want to assign a PO to multiple warehouses. A: Two separate Assign actions - one per warehouse. Each assignment records its own set of PO lines, so the two never overlap.

Q: Why can't I see prices in Tab 4 / on assignment cards? A: You CAN see prices in Tab 3 (Open POs drilldown). Tab 4 (Assign to Warehouse) shares the read with Warehouse users, so prices are hidden there for consistency.

Q: Force-closure undo? A: Not yet. Once force-closed, you'd have to either raise a Vendor Return (reopens the line) or admin-edit via Master DB Editor. A 24-hour undo window is on the v3.0 backlog.

16. Cross-Role Procurement Walk-through

A single happy-path narrative threading all five roles together, useful for onboarding.

16.1 The scenario

Site GI-PS01 needs 50 RL panels and 20 BL bricks. Total 3 working days from PR raise to physical receipt.

16.2 Hour by hour

When	Who	What
Day 1 - 8:00 AM	Site HOD	Opens HOD Portal -> Purchase Requests. Creates 2 PR lines (50 RL panels, 20 BL bricks) with PR Number 3000099999, WBS Number 4003951, Network 4003951-PROJ-A, Plant GI-PS01, Delivery_Date Day 3.
Day 1 - 8:15 AM	Site HOD	Opens the Submit PR(s) to Logistics Portal expander -> multi-selects PR 3000099999 -> Submit Selected to Logistics.
Day 1 - 8:15 AM	(auto)	A "PR submitted to Logistics" notification lands in the Logistics inbox with a red badge, plus a WhatsApp ping if that event is enabled.
Day 1 - 9:00 AM	Logistics	Sees PR 3000099999 in Incoming PRs queue. Drills in - 2 lines, RL + BL. Clicks Use this PR to create a PO.
Day 1 - 9:30 AM	Logistics	Switches to Create PO -> manual entry. Picks vendor "Carborundum Universal" (0000110341), Inco/Payment auto-fill, fills PO Number 4720033030, Unit_Price per line, Total_Price computed. Save PO.
Day 1 - 9:30 AM	(auto)	The PR lines are marked as covered by a PO and disappear from the Logistics queue. The site HOD gets a "PO issued" notification with the

When	Who	What
		vendor and PO number.
Day 1 - 10:00 AM	Logistics	Assign to Warehouse -> picks PO 4720033030 + Warehouse WH-A + Expected Delivery Day 2 evening. Assign.
Day 1 - 10:00 AM	(auto)	A "PO assigned to warehouse" notification lands in the WH-A users' inbox, plus a WhatsApp ping if that event is enabled.
Day 1 - 11:00 AM	Warehouse user (WH-A)	Incoming Assignments -> [OK] Acknowledge assignment #N. Goes about their day.
Day 2 - 5:00 PM	Warehouse user	Physical truck arrives from vendor. Receive Goods -> assignment #N -> type 50 in Receive Now for the RL line, 20 for the BL line. Record receipt. Both lines are marked delivered.
Day 2 - 5:30 PM	Warehouse user	Prepare DN -> pick PO 4720033030, destination GI-PS01. Types Ship Qty 50 for RL line, attempts to add BL line - rejected (RL/BL strict separation). Saves DN-WH-A-20260617-001 for RL only.
Day 2 - 5:35 PM	Warehouse user	Repeats: new DN with only the BL line -> DN-WH-A-20260617-002.
Day 2 - 5:40 PM	Warehouse user	Outbound DNs -> submits both drafts to Logistics.
Day 2 - 5:45 PM	Logistics	Sees DN-WH-A-20260617-001 + 002 in their DN approval queue. Confirms delivery date Day 3 AM. Approves both.
Day 2 - 5:45 PM	(auto)	Both DNs move to awaiting-HOD-approval. The site HOD gets the "DN approved by Logistics" notification.
Day 2 - 6:00 PM	Site HOD	Logs in, sees red bell badge 2 unread. Opens inbox -> drills into the DNs via DN Approvals tab. Approves both.

When	Who	What
Day 2 - 6:00 PM	(auto)	Both DNs move to awaiting-Store-Keeper, and a matching draft receipt appears in the site's Receipt Staging for each. The site SK gets the incoming-DN notification.
Day 3 - 8:30 AM	Truck arrives at site GI-PS01	
Day 3 - 8:45 AM	Site SK	Opens Entry Log -> Receipt Staging. The new Incoming Delivery Notes from Warehouse expander is open at the top, showing both DNs. Inspects both, confirms physical match, clicks [OK] Mark as Received on each.
Day 3 - 8:45 AM	(auto)	Two receipts are posted to the ledger, each carrying its DN number, source warehouse and originating PO. Both DNs are marked received, the staged drafts are cleared, and the Live Dashboard shows the new stock immediately.

16.3 Side-path examples

A site rejects a DN

Same scenario, but on Day 2 at 6:00 PM, Site HOD inspects the line preview on DN-002 and notices the BL qty doesn't match the PR. Rejects DN-002 with note "BL qty in DN is 25, PR asked for 20 - bounce back".

- DN-002 is marked rejected, and the reason is stored with it
- Warehouse user gets the rejection notification with the reason
- The BL line on the PO frees up qty (the over-ship guard recalculates available since DN-002 is no longer "live")
- Warehouse prepares DN-003 with correct qty

Logistics force-closes the PR mid-flight

Project cancelled by management on Day 1 evening:

- Logistics -> Force-Close -> radio = PR -> reason "Project cancelled per CFO email 2026-06-16" -> Force-close PR
- Admin + Site HOD see critical-severity notification in their bell
- The pending PO can either stay (if vendor already shipped) or also be force-closed

- Site HOD's In-Transit -> Force-closures sub-tab shows the closure with the reason

Warehouse asks for reschedule

Receiving day Truck doesn't show up - vendor delay:

- Warehouse -> Outbound DNs -> drill into DN-001 -> Request reschedule -> new date Day 4 -> reason "Vendor delivery delayed 24h confirmed by vendor email"
- Logistics -> Reschedules -> reviews -> [OK] Approve with notes "Confirmed with vendor"
- PO Expected_Delivery, po_assignments.Expected_Delivery, AND DN_Date all flip to Day 4 in one transaction
- Site HOD sees the decision notification

16.4 What stops a duplicate PR or a double order

Added 2026-08 (Phase 8). Three rules run underneath the procurement chain. None of them changes how the screens are used; they change what happens when two people, or one impatient person, do the same thing twice.

16.4.1 A PR number is reserved, not guessed

Every PR number is claimed in a registry before the requisition's lines are written, so two HODs pressing Create PR in the same second get different numbers. If a number is somehow taken, the system takes the next one rather than failing. Renaming a PR moves its reservation with it and is refused if the new name is already in use - by a registered PR or by one that was imported without ever being registered.

16.4.2 Each step happens once, in order

A requisition moves site_draft -> submitted -> in_po, and each move is checked against the state the row is actually in - not the state the screen was showing when it loaded. Consequences you will see:

- Submitting twice does nothing the second time. It used to accept lines that were already submitted, rewrite their timestamp and fire a second notification at Logistics.
- A PO cannot be raised twice over the same lines. The second attempt is refused with a message naming the lines, rather than silently succeeding.
- A PR may still carry several POs. Partial fulfilment splits one requisition across vendors or deliveries, so the lock is per line, not per PR.

16.4.3 A retry is not a second order

Submit PR, Create PO, Assign PO and Rename PR each carry a key minted when the form opens. Press the button twice, lose the connection and retry, or double-click - the system recognises the repeat and replays the first answer instead of doing the work again. Sending the same key with different details is refused, because that is a client bug rather than a retry.

[!] The buttons disappear rather than grey out. Once a PR is submitted, the Submit button is gone - not disabled. A greyed-out button invites a second click and a support call about why it "isn't working"; an

absent one says the step is done.

22. Quality, Safety, Employees & Procurement (QSEP)

Everything in this chapter went live in August 2026. It adds four things that used to live on paper or in somebody's memory: who checked the material, who is wearing the safety gear, where a person works, and what the scanned purchase paperwork actually said.

Read section 22.1 first even if only one of the four concerns you - the QC approval rule changes what happens when a Store Keeper tries to issue certain materials, and that affects the whole site.

22.1 Quality Control - the new role and the issue block

What a QC does

A Quality Control inspector checks material that carries risk if it is wrong. Today that means the Surface Shields category - 36 materials out of the 466 in the master list. Nothing else in the catalogue is affected by any rule in this section.

> ### The rule, in one line > > Material without an MTC CAN be sent and dispatched to site. It CANNOT be > issued or consumed at the site. > > Receiving it is fine. Putting it on a Delivery Note and driving it to site is > fine. Handing it to a worker is not. If you have been told otherwise, that > came from a document written before 2026-08-12 - the rule below is what the > system actually does.

There are two separate gates. Both apply at the same moment - the instant before material goes to a worker - but they are satisfied by different people, which is what makes them worth telling apart:

Gate	When it applies	What it demands	Who is stopped	Who fixes it
Material Test Certificate	When a Store Keeper issues material to the field	A certificate on file covering that material at that site	The Store Keeper at the issue form	Logistics (or the warehouse, or the SK)
QC approval	The same moment	A QC must have inspected and approved that quantity at that site	The Store Keeper at the issue form	The QC

Material may be received, and may travel to site, with neither. That is deliberate and was decided by the operator. Holding a truck at the gate for paperwork that is sitting in somebody's inbox costs more than it saves - and worse, refusing to record a delivery does not stop it arriving; it only stops the system knowing about it, so real material sits in a yard invisible to the stock report and to everyone planning around it. What material may not do is reach a worker's hands unchecked. So both gates bind at issue.

When a controlled item is received without a certificate, Logistics is notified automatically to chase it from the supplier. Nobody has to remember.

You should almost never upload the same certificate twice

The certificate follows the material down the chain. Whoever has it files it once:

- Logistics attaches it to the purchase order line - before the goods have even shipped, and without knowing which site will end up with them;
- the warehouse attaches it to the Delivery Note as the material leaves;
- the Store Keeper uploads it at the site, from the Receive form or from the banner on the Issue form.

Any one of those clears the material for issue at the destination site, and the issue form tells you which document cleared it and where it came from. A certificate filed for one site does not clear another - it attests to one batch from one production run.

If you find yourself uploading a second copy of a document the company already has, stop and report it. That is a fault, not a workflow.

Raising an inspection

An inspection is opened automatically when controlled material is received. The QC does not create it by hand. It appears in Quality -> Inspections with a status of pending.

Deciding an inspection

Open the inspection and enter the quantity you approve. The status follows from the number, not from a separate choice:

You approve	Status becomes
Nothing (zero)	Rejected
Less than the submitted quantity	Partially approved
The whole submitted quantity	Approved

Rejected material stays in stock. It is not automatically sent back to the vendor and it does not disappear. It sits where it is, unusable, and is blocked from issue until somebody decides what happens to it. That was an explicit instruction: an automatic vendor return removes the evidence before anyone has looked at it.

What the inspector sees, and the Return No (added August 2026)

The inspection screen now shows the material's name, with the SAP and material codes underneath it. Before this it showed only the codes, which meant judging a material by a number rather than by what it is.

If a Material Test Certificate is on file, the certificate number appears with an Open certificate link that opens the actual document - in the queue and again in the inspection dialog. Previously it only said that a certificate existed, with no way to read it, which meant approvals were being made against documents nobody had opened.

When an inspector rejects any quantity, the system issues a Return No such as QCR-20260813-41. It is announced to the inspector on the spot, listed against the inspection, and sent to both the site's Store Keeper and its Head of Department. The Store Keeper types it into Return Stock and the return form fills itself in - see §13.10.

> This does not change the rule above. The Return No is an invitation for a > person to raise a return; nothing moves, and nothing is sent back to a vendor, > until the Store Keeper posts it and the HOD approves it.

The daily missing-certificate alert (added August 2026)

Every morning the system checks for Surface Shield material that is in stock with no Material Test Certificate on file, and tells the people who can do something about it:

Where the material is	Who is told
In a warehouse	Logistics, the Warehouse User, and the warehouse's QC
At a site	That site's Store Keeper, HOD and QC, and Logistics

Logistics appears on both lists because they are the only people who can obtain the document from the supplier. One message per location lists everything outstanding there, rather than one message per material.

The alert repeats every morning until the certificate is uploaded, and then stops on its own. That repetition is intended: this is a standing condition, not a one-off event, and the material cannot be issued to anybody while it lasts.

What the Store Keeper sees when the block fires

The issue form refuses with a message that names the actual numbers, so nobody has to guess which of three situations they are in:

- No inspection at all - "no quality inspection exists for it at this site. It cannot be issued to the field until a QC has checked the material and its certificate."
- Inspected but nothing approved yet - names how many inspections are still pending.
- Partly approved and used up - "QC has approved 40 and 40 is already issued or staged, leaving 0 - not enough for 10. Ask the QC to inspect the remaining stock."

The clearance is pooled per material per site, not per individual pallet. What is counted against it is everything already issued or staged awaiting HOD approval, from the date of the first inspection onward - issues that predate quality control entirely are not counted, or the site would be frozen forever by its own history.

> This does not change the FEFO or over-issue behaviour. Those remain > warn-and-record, as they always have. The quality block is about whether the > material is fit to use, which is a different question from whether the shelf > count agrees.

Creating a QC account

Quality -> QC Accounts, available to an HOD, a Warehouse User or Logistics. You create a QC inside your own scope: an HOD creates one for their site, a Warehouse User for their warehouse. Set a site or a warehouse, never both.

Moving a QC to another site is a request, not an action - the HOD raises it and an Admin decides. A QC whose remit could be changed by the person whose work they inspect is not an independent check.

22.2 PPE - issued through the ordinary form

There is no separate "Issue PPE" page, and this is deliberate. Safety boots leave the store the same way a drum of resin does: through Entry Log -> Issue. The form recognises that the item is PPE and asks for the extra details on the spot.

Select a PPE item and three fields appear:

Field	When it is required	Why
Employee ID number	Always	PPE is tracked against a person, not a site
Safety Approval document	Unless a usable-time rule says otherwise	Somebody signed for this issue
Reason for early replacement	Only if the worker's current item has not expired yet	A boot replaced after three weeks is either a defect or a leak

The stock movement and the PPE record are written together. There is no second ledger and no separate PPE stock - the quantity leaves through the normal issue path, so stock levels, burn rate, reports and every other number stay correct without knowing anything about PPE.

The messages you will see

- "...is PPE - name the employee receiving it (their ID number), so it can be tracked against them."
- "worker 30816 is not in the employee master" - the person must exist on the roster first.
- "worker 30816 is at site RIYADH, not CNCEC - transfer them first if they have moved" - see 22.3.
- "Ahmed already holds this item, issued 2026-07-02 and good until 2026-10-02 - give a reason for replacing it early."

Usable time

Safety & People -> PPE Usable Time (Store Keeper or HOD) is where you say how long an item lasts: 90 days for gloves, 365 for a helmet, and so on. A rule can be global or set for one site, and the site's rule wins where both exist.

> [!] Until a rule exists for an item, two things follow. The issue will > always demand a Safety Approval document, and the item gets no expiry date > - which means it will never appear in the forecast below. If the PPE Forecast > looks empty, this is almost always why.

An expiry is a suggested replacement date, not a restriction. Nothing stops a worker using gear past it, no alert is sent to anybody's phone, and the worker is not blocked from anything.

The 15-day forecast

Safety & People -> PPE Forecast lists PPE expiring in the next fifteen days and tells you what to order:

> suggested = expiring - what is on the shelf - what is already on order

If the answer is zero because thirty units are already on an open purchase order, that is the correct answer and not a broken screen. The list shows the names of the people whose gear is expiring,

because a column of quantities cannot be sanity-checked by a human and a column of names can. A 90-day issue rate is shown beside the suggestion as a cross-check, never folded into it.

22.3 Employees - one person, one number

The employee ID number is the person. It is unique across the whole company, not per site, and everything else hangs off it.

Safety & People -> Employees shows the roster. An HOD can transfer someone to another site, and the transfer takes effect immediately - no approval step, because a person who has physically moved has already moved.

PPE history follows the person, not the site. Transfer a worker from one site to another and the boots they are wearing are still on their record. This falls out of keying on the ID number rather than being a feature bolted on top, which is why it cannot quietly stop working.

Admins get a timeline for any employee: every site they have worked at, with dates, and everything they currently hold. Use it when somebody asks "who had this, and when".

Employees -> Data Quality lists records that cannot be used and says why - a missing ID number, a duplicate, an inactive worker still holding gear. Check it before a PPE rollout rather than discovering the gaps one refused issue at a time.

22.4 Procurement - drafted delivery notes, urgency, and scanned paperwork

Delivery Notes draft themselves

When a warehouse receives goods against a purchase order, the system now prepares the Delivery Notes the sites are waiting for, grouped so that R/L and B/L material never share a note.

They are created as drafts. A person still adds the vehicle and driver and submits them, because the system cannot know that a truck exists. A notification tells the warehouse the drafts are waiting.

If a draft cannot be created - an over-shipment, a missing certificate - the goods receipt still stands. The stock genuinely arrived; losing that record because a convenience failed would be far worse. The reason is written to the audit log and the clerk cuts the note by hand.

Auto-drafting can be switched off in Settings.

Assigning a purchase order to a warehouse (changed August 2026)

A purchase order goes to one warehouse. The Purchase Orders list now shows a Warehouse column naming where each order was sent, and once an order has been assigned the Assign button is replaced by an assigned tag.

Previously the button stayed on every row for ever, and the list gave no indication that an order had already been routed. Pressing it a second time created a second assignment and told a second warehouse to expect the same goods, with nothing anywhere reporting a problem.

- Assigning the same warehouse again - a double-click, or an old tab - is accepted and changes nothing. No duplicate, no second notification.
- Assigning a different warehouse is refused, and the message names the warehouse that already holds the order. Re-routing an order another warehouse is already expecting is a decision for a person to make, not something the system will do quietly.

Shipping a delivery note (changed August 2026)

Pressing Ship now asks for two things before the delivery leaves:

- The delivery note number printed on the physical document - the carrier's or the supplier's number. This is not the DN number the system generated; the point is to connect the two, so that a delivery can be reconciled later without a telephone call.
- A scan or a photograph of the signed note. The camera button opens the phone camera directly, which is the practical option at a loading bay.

Both are required. If either is missing you are told which one, separately - one is typing and the other is scanning, and being sent to redo the half you had already done is its own small waste.

A Material Test Certificate is still optional here and works exactly as before. A certificate covers the material and is inherited from the purchase order or the delivery note down to the receiving site, so nobody uploads it twice. A delivery note covers this shipment and is inherited by nothing. They are different documents answering different questions.

Once shipped, the document number and a download link appear beside the delivery for everyone who touches it - the warehouse, Logistics, the HOD, Quality and the receiving Store Keeper. The Store Keeper is the one holding the paper copy while the truck waits, so this is where it matters most.

> Deliveries shipped before this change show "not shipped yet" in that > column. There is no document for them because none was ever captured, and > inventing a number would be worse than saying so.

Marking a reschedule urgent

A delivery reschedule can be marked urgent. An urgent one goes out immediately instead of waiting for the 4 p.m. evening digest. Use it sparingly - a channel where everything is urgent is a channel nobody reads.

Reading a scanned PR or PO

Upload the supplier's paperwork and the system reads the lines off it, so nobody retypes forty rows.

The file is stored before it is read, and this matters more than it sounds: the document that defeats the reader is precisely the one somebody will need to look at later. A failed read still leaves you the document.

There are two ways a document gets read, and the system chooses by whether the file actually contains text - not by whether it is a PDF:

The file	What happens
A PDF exported from a system	Text is read directly. Fast, exact.

The file	What happens
A PDF that is a photograph or a scan of a signed page	Read by the vision model. Takes longer; you poll for the result.

A printed-signed-and-scanned purchase order contains no text at all, only pictures of text. Before this change such a file was reported as read successfully with zero lines found - which is worse than an error, because nothing on screen distinguished "this order has no items" from "I could not read a word of it".

Once read, confirming the document links it to the purchase request or order you created, so the original scan is one click from the record. Offering a purchase order scan to a purchase request is refused by name - the two are not interchangeable.

> A quantity the reader could not make out is left blank, not zero. A zero on > a purchase order looks like an answer and would be ordered as one.

22.5 Assets - one hammer, one row, anywhere in the company

A serialised asset is now unique by its part number and serial number across the entire company, not per site. The same hammer cannot be registered twice by two sites who both think they have it.

Moving an asset between sites is a request approved by the HOD of the site it is leaving - the site giving something up is the one that needs to agree. On approval the asset's site changes and its old rack assignment is cleared, since a shelf in one yard means nothing in another. The move is recorded as a movement, so "where has this been" still answers correctly for the leg between the two sites.

Trying to re-register a serial that exists elsewhere is refused with a message that says where it actually is and what to do about it.

22.6 Passwords

The minimum password length is 8 characters, and a password must now also contain an uppercase letter, a number and a special character. This replaced a flat 12-character minimum with no composition requirement.

The rule is applied identically everywhere a password is set - admin creation, password reset and self-registration. Self-registration previously enforced a much weaker rule than the admin screens did.

The Request Access page now offers Quality Control in its role list.

26. Practice Mode - learn the system without touching Live

Practice is a separate copy of GI Hub for training. It has every page, every role and every workflow Live has, but it runs on its own made-up data in its own database. Nothing you do in Practice can reach Live: not a receipt, not an approval, not a message.

Use it to learn a job before doing it for real, to rehearse something you do rarely (a return, a stock count, a PR), or to try something out without worrying about the result.

26.1 Switching between Live and Practice

On the sign-in screen, above the username box, there is a Live | Practice switch.

- Live is selected unless you choose otherwise. It is the real system.
- Choose Practice and the page reloads, ready to sign in to Practice.
- Choose Live again to go back.

Things to know about switching:

- Switching signs you out. You sign in again in the environment you picked. This is deliberate: a browser is in one environment at a time.
- Your Live password does not work in Practice, and a Practice password does not work in Live. They are separate systems with separate accounts (§26.3).
- The installed apps (Windows, macOS, Android, iOS) have the same switch.

26.2 How you know you are in Practice

The page tells you in six ways, and none of them can be dismissed:

Where	What you see
Colours	Practice is violet: the sidebar, the header, the page background and the main buttons. Live is navy and gold.
Top left, under the GI Hub logo	A pulsing amber PRACTICE badge, on every page. On a phone it sits beside the menu button. On the sign-in page it appears at the top left as soon as you choose Practice. With Reduce motion on, it stays still.
Top of every page	An amber bar: PRACTICE - practice data, nothing here is real.
Page header	An amber PRACTICE tag beside the title, and an amber line under the header that stays when you scroll
Browser tab	The tab title starts with PRACTICE -

Where	What you see
Anything you print	A large diagonal PRACTICE - NOT REAL DATA watermark

These come from the server you are connected to, not from the switch. The server reports which environment it is, and the page shows what it says.

[!] If you see a red bar saying "Environment mismatch", the switch and the server disagree. Everything you try to save will be refused until that is fixed. Sign out, pick the environment again, and tell your administrator if the red bar comes back.

26.3 The practice accounts

Practice has one shared account per role. Your trainer gives you the password for these accounts.

Role	Practice account
Store Keeper	practice.storekeeper
Supervisor	practice.supervisor
Head of Department	practice.hod
Quality Control	practice.qc
Head of Qualities	practice.qchod
Warehouse	practice.warehouse
Logistics	practice.logistics
Auditor (view-only)	practice.auditor
Admin	practice.admin (separate password, held by your trainer or administrator)

The site-based accounts work at site CNCEC, and the Warehouse account works at warehouse WH-01.

Everyone in a class uses the same accounts, so you will see each other's practice entries. That is normal.

26.4 What is in Practice

The data is made up but realistic. The categories, units of measure, lining systems, work types and tank numbers are real. The material names, SAP codes, people, vendors, quantities, prices and dates are all invented. Nothing in Practice is a real person or a real stock figure.

It is the same data the training videos use. If a video shows Tutorial Demo Gasket Set (SAP 899001), you can find it in Practice.

Work is already waiting so every role has something to do: receipts, issues and a return waiting for HOD approval, a draft purchase request, and one already with Logistics.

Every new Live feature comes with a Practice example (rule 17g), so it can be tried before it is used for

real. The current ones:

Feature	Try it on
The store keeper's note filling the job card (§4.9a.2)	One tank and day with two notes, Floor - 12.5 SQM Done and Sump Wall - 4.2 SQM Done: two jobs, each with its area and part filled in.
Lots, FEFO and expiry (§3.10.4)	PRACTICE PU PRIMER (899971): PR-OLD (expired), PR-SOON (the FEFO lot), PR-LATE.
CHEMOLINE rolls (§3.10.4)	PRACTICE CHEMOLINE (899973), batch 1026009999, three rolls.
Lots used but never received (§3.10.3)	Lot PR-TYPO.
The assistant's quick answers (§3.12)	As practice.hod, ask the Hub Assistant "which lots expire in the next 30 days?" - a table, from Practice's own lots. As practice.storekeeper, ask "open the lots page" - an Open Lots & Expiry button - and "is there a video on staging a return?". Nothing new had to be added: the quick check reads whatever database the Practice process holds (rule 17).

26.5 What works differently in Practice

Everything works exactly as in Live, except these four things. Each one is switched off on purpose:

In Practice...	Why
No WhatsApp message or email is ever sent. Notifications still appear in the bell.	A practice alert must never reach a real phone or inbox.
Photo reading (OCR) is switched off. The page explains this, and you can still paste text or type rows in.	The AI engine is shared with Live and takes minutes per page, so trainees must never queue ahead of real forms.
Training videos watched in Practice are not counted. You can watch as much as you like, but I have watched and understood this is refused.	Training records are kept in Live only. Sign in to Live to have a video counted.
Two-factor authentication is off, and cannot be turned on.	The accounts are shared, so one person's authenticator would lock everyone else out.

26.6 Offline entries stay where they were made

If you lose signal while saving an entry, GI Hub keeps it and sends it when the connection returns (§3.7). In Practice this still works, with one extra guarantee: an entry saved in Practice is only ever sent to Practice. If you switch to Live before it has been sent, it waits, and it is sent the next time this browser is back in Practice. It can never be sent to Live, even if you sign in to Live on the same device.

26.7 Resetting Practice (admins)

Over time Practice fills up with everyone's practice work. An admin can put it back to its starting state:

1. Sign in to Practice as practice.admin.
2. Open Admin Console -> Practice. The page shows when the data was built and when, and by whom, it was last reset.
3. Click Reset Practice data... and type RESET PRACTICE DATA to confirm.

The reset takes about a second. It removes everything anyone has done in Practice, and everyone signed in to Practice is signed out. The approval queues are filled up again. Live is not affected in any way. It is a different database, and the Practice server has no permission to open it.

The Practice tab and the reset only exist in Practice. In Live there is no such button, and the server has no such function.

How administrators set up and run Practice on a server is in §17.9.